

**Enumeration of Springs in Buggavanka Watershed, Kadapa, Andhra Pradesh, India.***** Dr.T. Siva Prathap, \$ Shaik Asiya, #Dr.Mohd Akhter Ali, @M. Kamraju**

*#Assistant Professor, Research Scholar, \$ M.Sc student

* \$ Department of Earth Sciences, #@Department of Geography

*\$ Yogi Vemana University, #Osmania University, @Centre for Economic and Social Studies
Hyderabad, Telangana**Abstract**

*“Water” is one of the five basic elements of life out of **pancha bhuta** i.e. air, water, earth, fire and sky on the earth is the most essential element of life. Its availability is an essential component in socioeconomic development and poverty reduction (UNESCO-WWAP 2006). The demand for water is increasing in geometric progression all along the population growth (Gupta and Deshpande, 2004). Water on Earth goes through a gigantic Hydrological cycle of conversion from liquid to gaseous form by evaporation and coming down again to Earth as rainfall. Such water in that cycle appears as spring, is one of the important components for a many surface waters. These springs in that appear form a stream on such stream in the present study area is called Buggavanka. “Buggavanka” is a seasonal tributary to Pennar river basin, one of the major south Indian rivers. “Bugga” stands for “Spring” and “Vanka” for “Stream” in vernacular language Telugu.*

Classic definition of spring according to Meinzer (1923- 48) is “A spring is a place where, without the agency of man, water flows from a rock or soil upon the land or into a body of surface water”. In the present paper an effort is made to enumerate the existing, near to extinct and extinct springs within the watershed of the Buggavanka stream, in Kadapa, Andhra Pradesh, India. Geospatial technology formed the basis for the geotagging of the identified and explored springs in the study area, the tools employed in geospatial data processing made the study in a reconnaissance.

Watershed is a geohydrological unit where the water drains into a common single outlet. Such watershed boundary is delineated around the Buggavanka stream with an effort to bring the study with ecological, demographic and holistic approaches. The main idea of this study is to geotag the spring locations and create the database for future studies to augment the water discharge in the yet to die springs in watershed management perspective. The field visits and survey revealed many historical and cultural but are abstract in nature. However it emphasized for the detailed geohydrological study and participatory approach to bring back the glory of water flow as per the people living within the Buggavanka watershed.

Keywords: Spring, Watershed, Climate Change, Geospatial Technology

INTRODUCTION

“Water” is one of the five basic elements of life out of **pancha bhuta** i.e. air, water, earth, fire and sky on the earth is the most essential element of life. Its availability is an essential component in

socioeconomic development and poverty reduction (UNESCO-WWAP 2006). Water on Earth goes through a gigantic Hydrological cycle of conversion from liquid to gaseous form by evaporation and coming down again to Earth as rainfall. The demand for water is increasing in geometric progression all along the population growth (Gupta and Deshpande, 2004). Sustainable development is need of the hour and watershed forms the basic unit in water resource planning particularly in Semi Arid Tropics. Many drainages in semiarid areas and most desert drainages are ephemeral, containing running water seasonally, in response to rainfall, and not necessarily every year. “Buggavanka” is a seasonal tributary to Pennar river basin. “Bugga” stands for “Spring” and “Vanka” for “Stream” in vernacular language Telugu.

Watershed is defined as a geohydrological unit where all of its water gets collected at one point called as outlet. Classic definition of spring according to Meinzer (1923- 48) is “A spring is a place where, without the agency of man, water flows from a rock or soil upon the land or into a body of surface water”. In Buggavanka watershed all of the water both runoff and spring water joins with “Buggavanka” stream.

When the rain arrives there is a series of actions followed by rain water after reaching the ground. First of all it saturates the ground and then it will flow on the surface of the land as runoff. Then recharge occurs as the tertiary action at the recharge area whenever the water is stable or runoff velocity is slow. This groundwater from the aquifer will be discharged as springs, seepages, wells, bore wells etc. Springs are important sources of water to streams and other surface-water features. They are also important cultural and aesthetic features. Springs typically are present where the water table intersects the land surface. The water sources of such springs, in most cases are unconfined aquifers where the water flows under gravity and the recharge zone of such aquifer is vast throughout its elevated area. In fractured-rock terrain, springs are fed through faults or fractures. The water sources of such springs are confined aquifers. (Technical Guide to Managing Ground Water Resources, May 2007, USDA).The recharge zone of those springs is the outcrop of the source



aquifers and also throughout its elevated area as the confining strata above aquifer is fractured, it supports recharge. For the aquifers with massive confining strata above that out crop is the only recharge zone.

In 2001 Fetter identified five types of springs. They are

1. Depression springs:-At low topographic spots where the water table reaches the surface.
2. Contact springs: - At a lithologic contact between rock of contrasting permeability.
3. Fault springs: - At Faulting zones which can form a boundary to ground water flow.
4. Sinkhole springs: - If the artesian pressure in the subterranean solution cavities is high enough to reach the surface, a sinkhole spring is formed.
5. Fracture springs: - ground water flowing along a fracture or joint intersects the land surface.

Spring water discharge fluctuations owe primarily due to rainfall pattern in the recharge area or more precisely stated, to variation in the amount of rainwater that is able to infiltrate the ground and recharge the ground water. Marked variation in the discharge following rainfall indicates rapid infiltration of rainwater and recharge of the groundwater, it show strongly periodic seasonal rhythm. Superimposed on these variations is a periodic (monthly) fluctuation resulting from occasional heavy rain falls, generally in the rainy season. Following rain, water starts to percolate down through the soil stratum (Technical Guide to Managing Ground Water Resources, May 2007, USDA). The increased recharge of groundwater augments the discharge of the springs and seepages, and thus, causes more rapid outflow of the ground water following the augmentation of duration of river flow providing water for appropriate purposes at least for some more time in the year.

Geospatial Technology

Remote sensing is the process of acquiring information about an area from a distance without directly establishing a physical contact with that area. Satellites, aircrafts and other ground borne platforms are used in remote sensing. The collected information is in the form of imagery, photographs. The maps prepared from these sources are opened in GIS soft ware and used for locating and analyzing target areas, drawing the boundaries etc.

Digital image processing is an important aid to process satellite images. Image preprocessing, image processing, image rectification etc. can be

done using some digital image processing software such as ERDAS IMAGINE, ILWIS etc. Activities such as layer stacking, noise correction, image enhancement, classification are some of the examples performed by this software.

REVIEW OF LITERATURE

A brief review of previous work done on various aspects of watershed management is presented.

Reddy and Walker (1987).from their study on "impact of watershed programmers on adoption and economics of technology and on economic conditions of rural people" reported 16 per cent of arm under sole crops in Mitteniari watershed area as against 12 per cent in non-watershed villages. It is also observed that the area covered under improved varieties in watershed villages was 92 per cent as against 69 per cent in non-watershed area villages during the Kharif 1986.

of groundwater of 200 km³ , out of which 100 km³ most likely occurs in Western India.

Ratna Reddy, Y.R. (2000) observed that watershed development programme had brought fortunes for the rural development in India by improving the socio-economic status of the rural people.

Narashiman, T.N. (2008) suggested that ground water would be best managed cooperatively through local user groups and panchayathraj institutions with technical inputs from the groundwater boards at central and state level. Artificial recharge and rainwater harvesting should be actively encouraged through the use of modern methods.

OBJECTIVES

1. To Geotag the springs within the Buggavanka watershed area.
2. To enumerate the springs for future studies.

STUDY AREA DESCRIPTION

The study area falls under Semi Arid Tropics in India which is characterized by varied rainfall in time and space under the influence of the vagaries of monsoons. The area is situated south of Kadapa district in Andhra Pradesh including the parts of nine mandals. They are The Buggavanka watershed area is 724.73Km² and located between 14°12'15"N & 14°33'16"N Latitudes and 78°38'17"E & 78°57'52"E Longitudes. The highest relief is 818 meters which is associated with Palakonda Hills i.e., upper reach in the south and the lowest 119 meters at the confluence of Pennar river in the north.



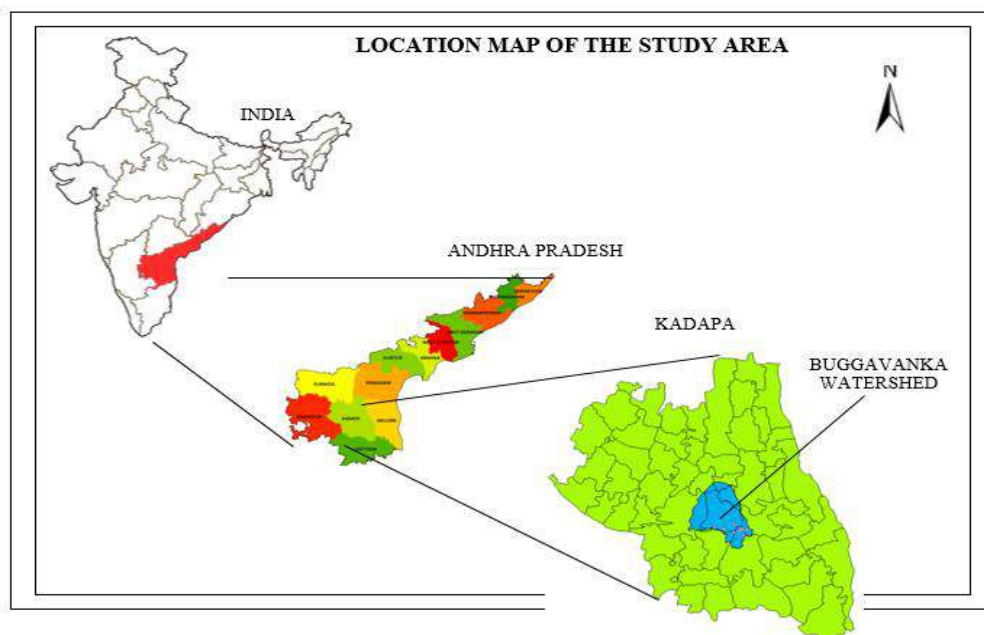
This area receives rainfall from SW Monsoon. September is the rainiest month. July and august months also receive excess rain. In response of this rainfall the spring will become active and the river flows. However it receives winter rainfall under the influence of cyclonic activity in Bay of Bengal during the post monsoon season. The southern portion of the study area is hilly terrain. The climate is medium to dry with mean annual temperature of 32°C.

Buggavanka watershed is covered in nine mandals of Kadapa district in Andhra Pradesh. They are Chintakomma dinne, Pendlimarri, Kadapa, Veeraballi, Vontimitta, Chennur, Ramapuram, Vallur and Sidhout. And a total of 54 villages where the majority of the stake holders are from the Chintakommadinne mandal. The details of area of watershed covered mandal wise and the number of villages sharing the area in each mandal and number of villages becoming the part of the watershed are mentioned hereunder for ready reference in the Table.1 as follows.

Table 1. Mandals in Buggavanka watershed

Sl.No.	Name of the Mandal	Area of the Mandal covered in Buggavanka watershed (Sq km)	Number of villages of the mandal covered in Buggavanka watershed
1.	Chintakomma dinne	333.94	23
2.	Pendlimarri	130.10	5
3.	Kadapa	84.94	11
4.	Veeraballi	63.79	
5.	Vontimitta	38.93	1
6.	Chennur	28.77	8
7.	Ramapuram	27.39	
8.	Vallur	17.05	6
9.	Sidhout	0.308	
TOTAL		724.73Km ²	54

Fig. 1 Location Map





The Buggavanka stream traverses many villages within watershed providing drinking water and irrigation facilities either directly or indirectly. The major part of Kadapa town depends on this stream as means of domestic and drinking water supply by the local governing bodies through pump houses in its riparian part. Buggavanka dam is present across this tributary in Kadapa, which is a medium irrigation project.

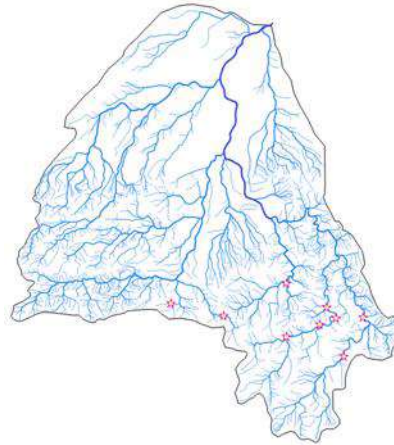


Fig . 2 Buggavanka Drainage

The Buggavanka stream is a tributary to Penna River. The stream headways are in Palakonda Hills and the mainstream is about 57.947 Km long. It is joined by seven small tributaries finally to feed Pennar River Basin. As the Palakonda Hills are at the southern margin of Cuddapah basin, they are structurally disturbed hill ranges. Hence they possess so many ground water springs besides beautiful waterfalls. With the dense natural flora and vast fauna, they flourish during the rainy season. The springs of these hills that are present in the Buggavanka watershed contribute to Buggavanka stream. The watershed is covered in SOI toposheets 57J10, 57J11, 57J14, 57J15 and 57J16 on 1:50000 scale. But springs are covered in 57J15 (D44G15) toposheet only. The springs are marked with “blue colored plus symbol” on SOI Toposheets. By observing this, 8 springs are identified within the delineated boundary. They are :

Fig. 3 Topomosaic

1. Masimadugu
2. Pakalagundam
3. Ankumadugu
4. Kortimadugu
5. Muluvedurumadugu
6. Jalarevula spring
7. Chitralamadugu
8. Unnamed Spring



FIELD SURVEY

Field visits need to be conducted at the locations of spring. It is essential that flow from a spring is identified as to the geologic formation from which the water discharges. The following are among the basic data collection requirements for springs. (Technical Guide to Managing Ground Water Resources, USDA)

1. Latitude-longitude values at spring location
2. Elevation of the spring.
3. Uses of the spring water.
4. Permanence of flow.
5. Discharge pattern of the spring in the year
6. Chemical characteristics of the water.



7. Temperature of spring water
8. Type of spring.
9. Source aquifer.

As there is no any available previous field data about spring their locations are not known. To know their Latitude, Longitude and Elevational values, the springs shape file is converted to kml file in QGIS and then it is viewed in Google Earth for Navigation. The springs in the Buggavanka watershed area are physically verified and location is geotagged for future studies. The detailed information is collected from the local villagers as inventory method for ready reference.

Fig. 4 Field Photographs



Spring water discharge fluctuations owe primarily due to rainfall pattern in the recharge area or more precisely stated, to variation in the amount of rainwater that is able to infiltrate the ground and recharge the ground water. Marked variation in the discharge following rainfall indicates rapid infiltration of rainwater and recharge of the groundwater, it show strongly periodic seasonal rhythm. Superimposed on these variations is a periodic (monthly) fluctuation resulting from occasional heavy rain falls, generally in the rainy season. The last ten years monthly average rainfall of Kadapa district is presented as below which follows the flow pattern of Buggavanka stream.

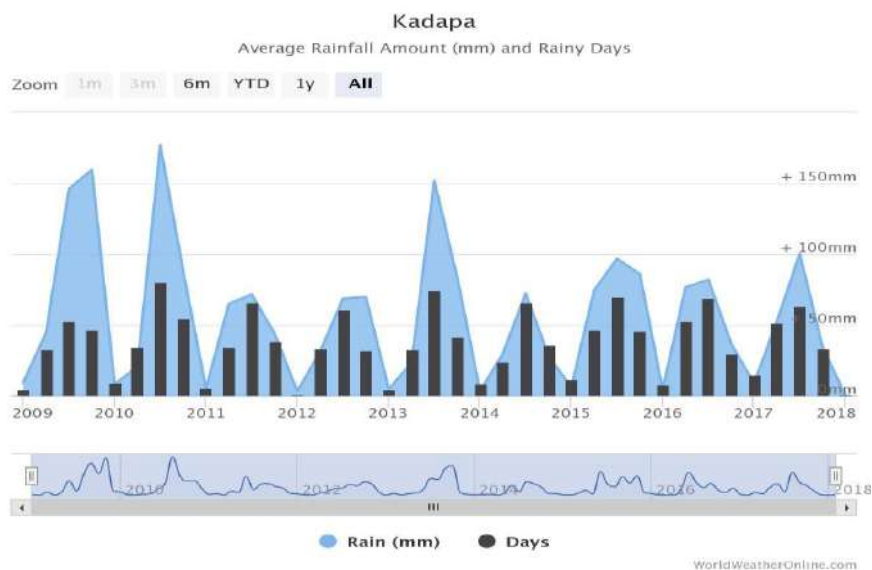


Fig. 5 Kadapa Rainfall Graph



The average annual rainfall of Kadapa is 710 mm. From the data presented in figure 5 it is clearly evident that there is a change in rain fall pattern indicating the climate change impacts. The cumulative annual rainfall of the years 2009 and 2010 is 1079.54 and 881 respectively. It is more than the average and sufficient to activate the springs. Hence the springs activated for full range during those years and the streams discharged well. There is a sudden decrease in rainfall in the succeeding years 2011 and 2012 that is 551.83, 519.58 respectively. It is very less than average rainfall. Hence during those years the stream flow is very less. 2014 is year which received the poorest rainfall among these representing a drought year. The years 2013 and 2015 received appreciable rainfall 790.54 and 790.5 respectively. Again there is decreasing trend in rainfall during the years 2016 and 2017 that is 599.16, 588 respectively. During these years the springs are activated but they were not successful to feed the stream. The water reached the dam and then there is no appreciable flow after

crossing that. This data is plotted in the graph. The days in each month received rainfall are plotted as histogram and the rainfall per month is received in the form of curve.

SUMMARY

The main theme of this study is to identify the existing active and inactive springs within the Buggavanka watershed area. The identified springs were classified as active and inactive under the influence of Rainfall and climate change. The springs are geotagged and the discharge summary is well gathered from the local villagers for ready reference. With the integration of geohydrological studies and enquiry of the history of the spring discharge in the recent past, it is taken for detailed study as the scope of rejuvenation is promising. The geohydrological studies amalgamated with the Rainwater harvesting techniques along with biological and social measures are considered for the future studies on springs of Buggavanka watershed.

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